

Project on Freestyle Jobs

To Begin with the Lab

Summary of the Lab

This lab demonstrates deploying a Java web application using **Jenkins Freestyle Jobs** with two EC2 instances: a **Jenkins server** and a **Tomcat server**. The Tomcat server is configured by installing **Java (Amazon Corretto 1.8)** and **Apache Tomcat**, and verifying it through port **8080**. On the Jenkins server, the **Publish Over SSH plugin** and **Maven** are configured. A Freestyle job is created that pulls source code from **GitHub**, builds the project using **Maven (compile and package)**, and generates a WAR file. Jenkins then transfers the WAR file to the Tomcat server using SSH, deploys it in the **webapps** directory, and the application is accessed via the Tomcat URL.

- EC2 Instance (Two servers) are used:
- Jenkins Server (Current EC2 Instance):
 - Maven, Publish Over SSH plugin
- Tomcat Server (Create a new EC2 Instance):
 - Java, Apache Tomcat
- Create a new EC2 instance with the name Tomcat Server.
- Recommended configuration:
- Instance type:
 - t2.medium
- Storage:
 - 15 GB
- Security Group:
 - Allow HTTP (80),
 - Allow HTTPS (443),
 - Allow 8080, Allow SSH (22)
- Launch the instance.
- Connect to the server using SSH.
- [Install Java:](#)

```
sudo yum install java-1.8.0-amazon-corretto-devel
```

```
[ec2-user@ip-172-31-23-184 ~]$ sudo yum install java-1.8.0-amazon-corretto-devel
Amazon Linux 2023 Kernel Livepatch repository                               245 kB/s | 30 kB   00:00
Dependencies resolved.
-----
Package                                Architecture      Version                               Repository         Size
-----
Installing:
java-1.8.0-amazon-corretto-devel       x86_64            1:1.8.0_482.b08-1.amzn2023          amazonlinux        63 M
Installing dependencies:
adwaita-cursor-theme                   noarch            47.0-1.amzn2023.0.1                 amazonlinux        325 k
```

- Verify:

```
java -version
```

```
[ec2-user@ip-172-31-23-184 ~]$ java -version
openjdk version "1.8.0_482"
OpenJDK Runtime Environment Corretto-8.482.08.1 (build 1.8.0_482-b08)
OpenJDK 64-Bit Server VM Corretto-8.482.08.1 (build 25.482-b08, mixed mode)
[ec2-user@ip-172-31-23-184 ~]$
```

- Install Apache Tomcat
- [Download Tomcat:](#)

Binary Distributions

- Core:
 - [zip \(pgp, sha512\)](#)
 - [tar.gz \(pgp, sha512\)](#)
 - [32-bit Windows zip \(pgp, sha512\)](#)

```
wget https://dlcdn.apache.org/tomcat/tomcat-9/v9.0.118/bin/apache-tomcat-9.0.118.tar.gz
```

- Extract the package:

```
tar -xvzf apache-tomcat-9.0.118.tar.gz
```

- Move inside Tomcat directory:

```
cd apache-tomcat-9.0.118/bin
```

- Look for the startup.sh file

```
[ec2-user@ip-172-31-23-184 ~]$ cd apache-tomcat-9.0.118/bin/
[ec2-user@ip-172-31-23-184 bin]$ ls
bootstrap.jar      catalina.sh        commons-daemon-native.tar.gz  configtest.sh  digest.sh  setclasspath.bat  shutdown.sh  tomcat-juli.jar  tool-wrapper.sh
catalina-tasks.xml  cipher.bat        commons-daemon.jar           daemon.sh       makebase.bat  setclasspath.sh  startup.bat  tomcat-native.tar.gz  version.bat
catalina.bat       elphers.sh        configtest.bat              digest.bat      makebase.sh   shutdown.bat     startup.sh  tool-wrapper.bat   version.sh
[ec2-user@ip-172-31-23-184 bin]$
```

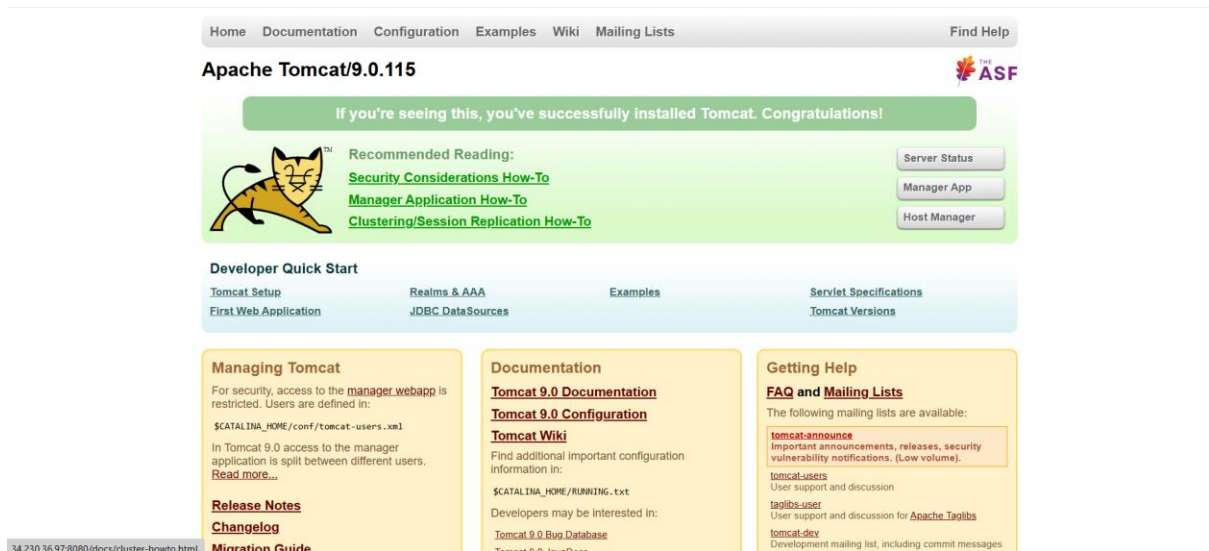
- Start Tomcat:

```
./startup.sh
```

```
[ec2-user@ip-172-31-23-184 bin]$ ./startup.sh
Using CATALINA_BASE:   /home/ec2-user/apache-tomcat-9.0.115
Using CATALINA_HOME:   /home/ec2-user/apache-tomcat-9.0.115
Using CATALINA_TMPDIR: /home/ec2-user/apache-tomcat-9.0.115/temp
Using JRE_HOME:        /usr
Using CLASSPATH:        /home/ec2-user/apache-tomcat-9.0.115/bin/bootstrap.jar:/home/ec2-user/apache-tomcat-9.0.115/bin/tomcat-juli.jar
Tomcat started.
[ec2-user@ip-172-31-23-184 bin]$
```

- Check Tomcat in browser:

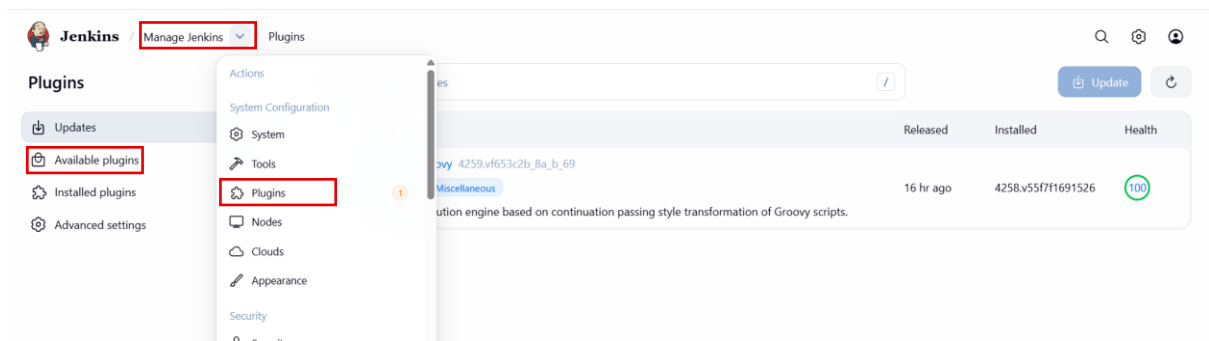
```
http://<TOMCAT_PUBLIC_IP>:8080
```



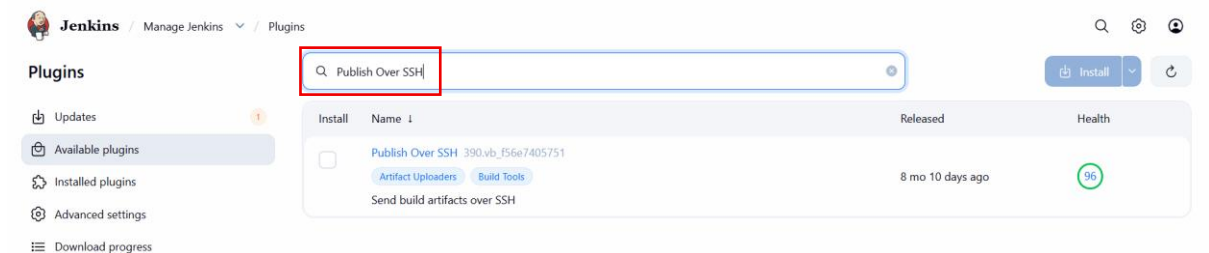
- If Tomcat page appears → installation successful.
- Stop Tomcat (for deployment):

```
./shutdown.sh
```

- Install Jenkins Plugin (Publish Over SSH)
- Go to Jenkins Dashboard
- Manage Jenkins
- Plugins
- Available Plugins

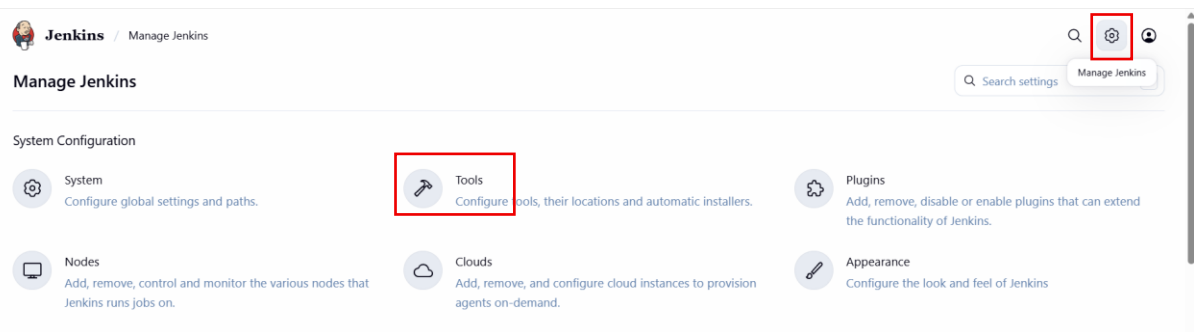


- Search: Publish Over SSH

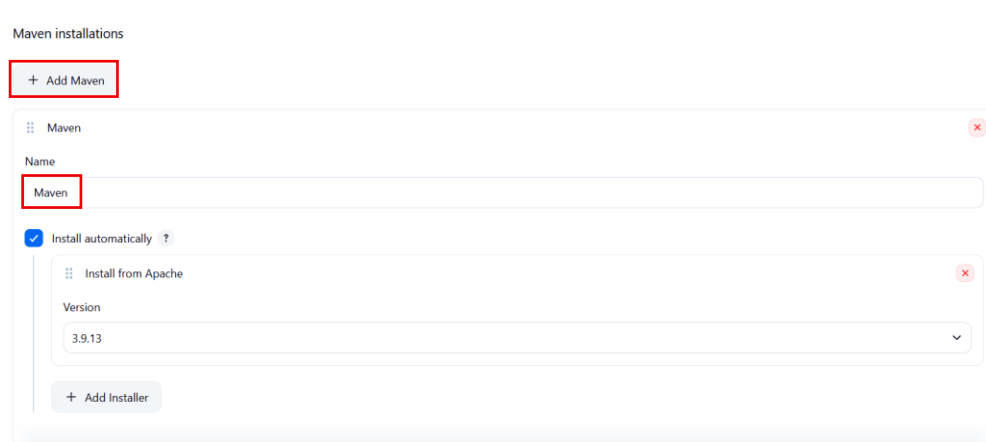


- Install the plugin.
- Open **Jenkins Dashboard**.
- Click **Manage Jenkins**.

- Click **Tool**.



- Scroll to **Maven Installation**.
- Click **Add Maven**.



- Click **Save**.
- If you have already done with part you can skip.
- Go to Jenkins Dashboard.
- Click **New Item**.

New Item

Enter an item name

Maven-Freestyle

Select an item type



Pipeline

Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.



Freestyle project

Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.



Multi-configuration project

Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.



Folder

Creates a container that stores nested items in it. Useful for grouping things together. Unlike view, which is just a filter, a folder creates a separate namespace, so you can have multiple things of the same name as long as they are in different folders.

OK

The image shows the Jenkins configuration page for a new item named 'maven-freestyle'. The left sidebar contains a 'Configure' menu with options: General, Source Code Management, Triggers, Environment, Build Steps, and Post-build Actions. The 'General' tab is selected. The main configuration area includes a 'Description' field with the text 'Addressbook application', a 'Plain text' / 'Preview' toggle, a checked 'Discard old builds' checkbox, a 'Strategy' dropdown menu, a 'Log Rotation' dropdown menu, a 'Days to keep builds' input field with the value '30', and a 'Max # of builds to keep' input field with the value '1'.

- Navigate Source Code Management
- Select the git option from there

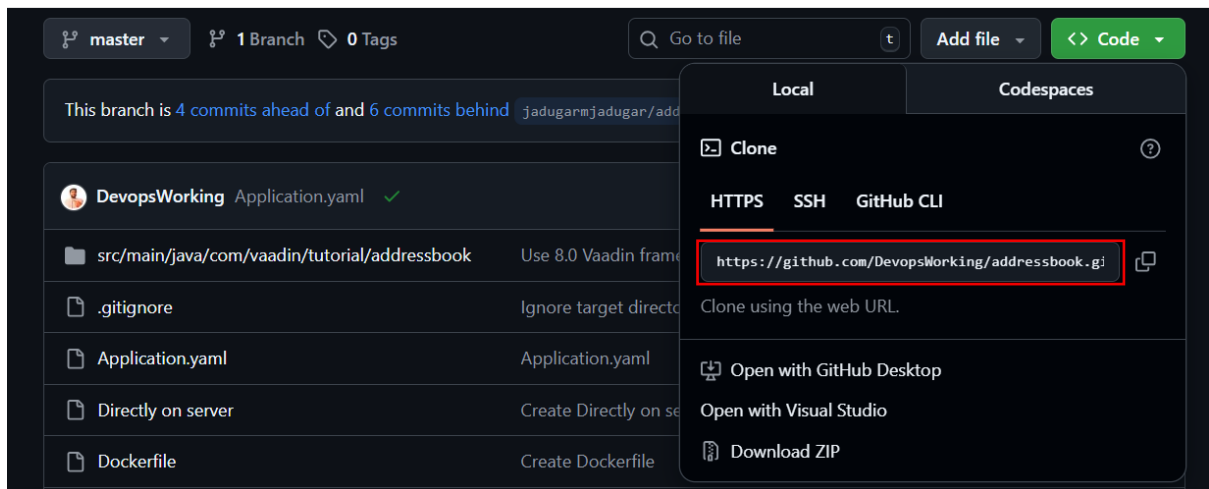
Source Code Management

Connect and manage your code repository to automatically pull the latest code for your builds.

☐ None

☒ Git ?

- After selecting git.
- Copy the URL from the github



- Enter the Repository URL
- Note: If your repository is private, you have to select the credentials form the credential option.

Repositories ?

Repository URL ?

`https://github.com/DevopsWorking/addressbook.git`

! Please enter Git repository.

Credentials ?

- none -

+ Add

Advanced ▾

- Check you github repository is master or main.

Branches to build ?

Branch Specifier (blank for 'any') ?

`*/master`

Environment

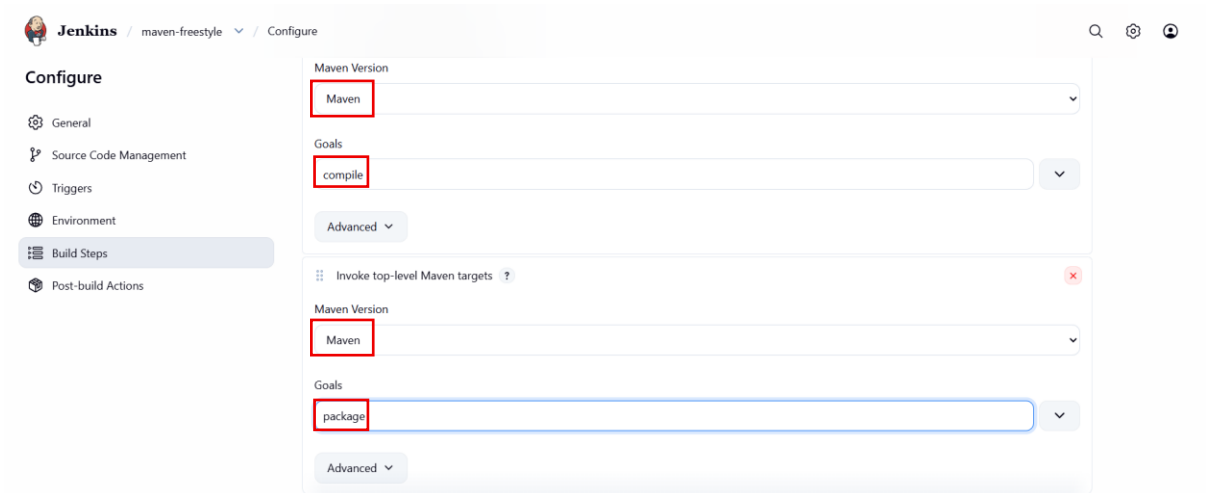
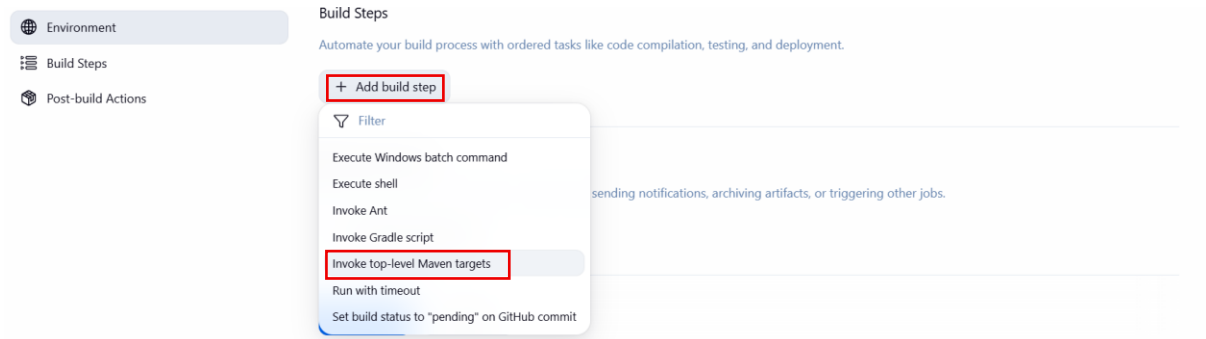
Configure settings and variables that define the context in which your build runs, like credentials, paths, and global parameters.

☒ Delete workspace before build starts

Advanced ▾

- ☐ Use secret text(s) or file(s) ?
- ☐ Send files or execute commands over SSH before the build starts ?
- ☐ Send files or execute commands over SSH after the build runs ?
- ☐ Add timestamps to the Console Output
- ☐ Inspect build log for published build scans

- Navigate to the Build Step section
- Click on the Add Build Step
- Select the 'Invoke top-level Maven targets'



- Similarly make Goals for – compile and package
- Save the Job
- Now click on the Build now
- After saving it will create a job successfully
- Configure Publish Over SSH
- Go to:
 - Manage Jenkins
- Go to System
- Scroll to:
 - Publish Over SSH
- Click Add Server.
- Fill details:
- Name:
 - tomcat-server
- Hostname:
 - <Tomcat_Public_IP>

- Username:
 - ec2-user
- Remote Directory:
 - /home/ec2-user/apache-tomcat/webapps

Jenkins Manage Jenkins System

SSH Server

Name ? tomcat-server

Hostname ? 34.230.36.97

Username ? ec2-user

Remote Directory ? /home/ec2-user/apache-tomcat-9.0.115/webapps

☐ Avoid sending files that have not changed ?

Advanced ▾

```
[ec2-user@ip-172-31-23-184 bin]$ cd ..
[ec2-user@ip-172-31-23-184 apache-tomcat-9.0.115]$ cd ..
[ec2-user@ip-172-31-23-184 ~]$ cd apache-tomcat-9.0.115/
[ec2-user@ip-172-31-23-184 apache-tomcat-9.0.115]$ ls
BUILDING.txt CONTRIBUTING.md LICENSE NOTICE README.md RELEASE-NOTES RUNNING.txt bin conf lib logs temp webapps work
[ec2-user@ip-172-31-23-184 apache-tomcat-9.0.115]$ cd w
webapps/ work/
[ec2-user@ip-172-31-23-184 apache-tomcat-9.0.115]$ cd webapps/
[ec2-user@ip-172-31-23-184 webapps]$ pwd
/home/ec2-user/apache-tomcat-9.0.115/webapps
[ec2-user@ip-172-31-23-184 webapps]$
```

- In Advanced section:
 - Use password authentication or use a different key
- Add Private Key
- Paste the EC2 private key (.pem file).

Advanced ^ Edited

☒ Use password authentication, or use a different key ?

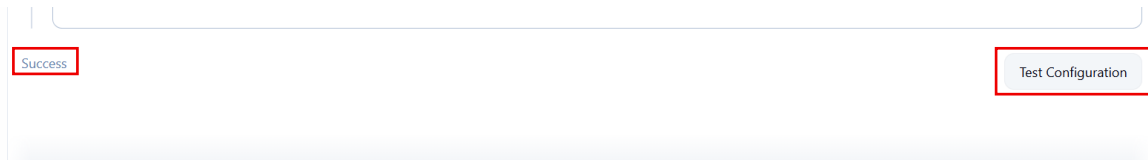
Passphrase / Password ?

Path to key ?

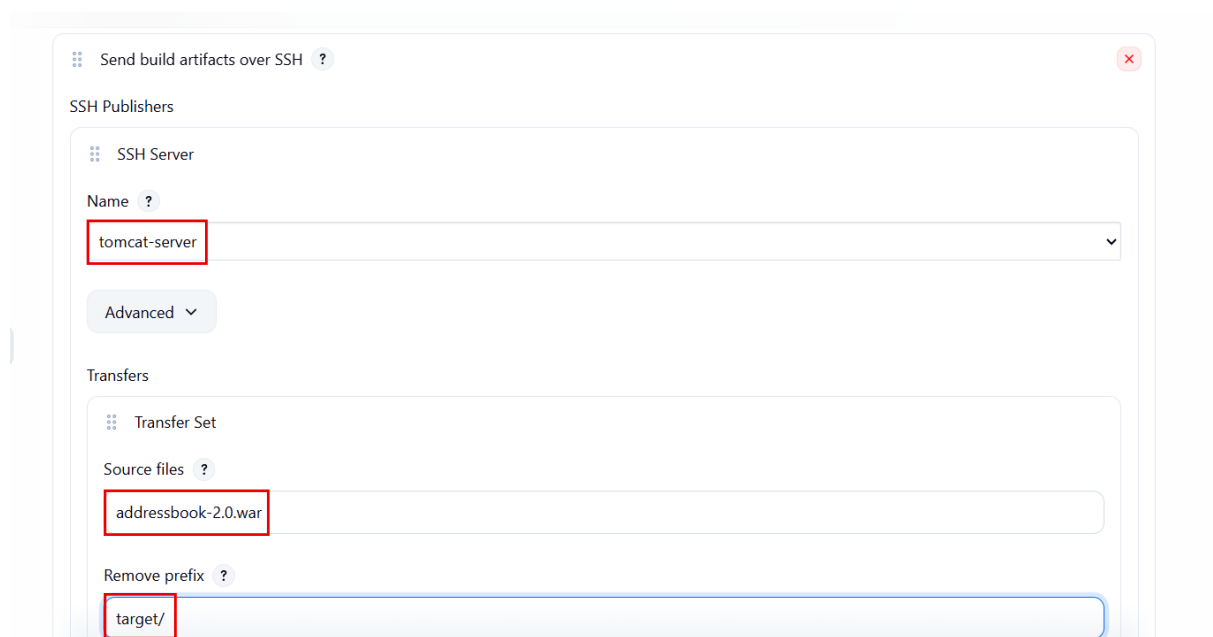
Key ?

```
6XBIDcfHfp0hiaC0AjuWVgC95Jf8EWLVcb90wge/Kvijxs3Tcy/WRgX6Pxrppf
QvpyqtCh3d1EGrCPdrljsZUOsaHUQwgekzPNRGLs37w6nN1P3MeC8c4/D8GaEud
LsxRJqUCgYEAreXjUCRioJTJOo+iY9wC2ICpd9O+IVCsYnjDgyfNU7+nroFnW1
085AHYIYLQvHZBZVoR+EEtaGetRHhBAGwWavxt73czdFS4Kjue0P3/QjUnTcEI2
o3FfAxwDULksfQLMBR83njBbqTqrKfs0Pv10H6hL+znzgcIWZRZ/9Y=
-----END RSA PRIVATE KEY-----
```

- Click:
 - Test Configuration
- If configuration is correct → Success message appears.



- Click Save.
- Open Jenkins job > Click Configure
- Configure Artifact Transfer
- Look for **Post-build Actions**
- Select:
 - Send build artifacts over SSH
- Choose server:
 - Tomcat-Server
- Provide source file:
 - addressbook-2.0.war
- Remove prefix:
 - target/



- Click Save.
- Click on Build Now
- In console Output shows:
 - Transferred 1 file

```
SSH: Connecting from host [ip-172-31-13-72.ec2.internal]
SSH: Connecting with configuration [tomcat-server] ...
SSH: Disconnecting configuration [tomcat-server] ...
SSH: Transferred 1 file(s)
Finished: SUCCESS
```

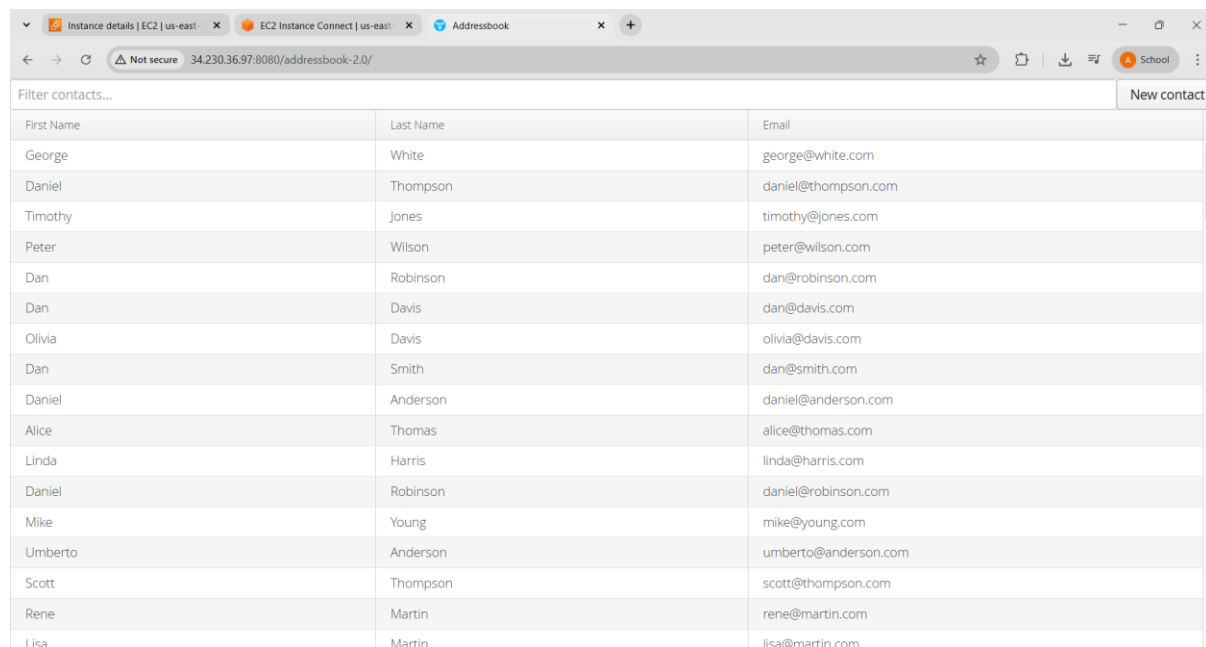
- Verify WAR File on Tomcat Server
- Login to Tomcat server:
 - ls
- You should see:
 - addressbook-2.0.war

```
[ec2-user@ip-172-31-23-184 webapps]$ ls
ROOT addressbook-2.0.war docs examples host-manager manager
[ec2-user@ip-172-31-23-184 webapps]$ ls -ll
total 13836
drwxr-x---. 3 ec2-user ec2-user 16384 Mar 12 14:52 ROOT
-rw-rw-r--. 1 ec2-user ec2-user 14133230 Mar 12 16:48 addressbook-2.0.war
drwxr-x---. 16 ec2-user ec2-user 16384 Mar 12 14:52 docs
drwxr-x---. 7 ec2-user ec2-user 99 Mar 12 14:52 examples
drwxr-x---. 6 ec2-user ec2-user 79 Mar 12 14:52 host-manager
drwxr-x---. 6 ec2-user ec2-user 114 Mar 12 14:52 manager
[ec2-user@ip-172-31-23-184 webapps]$ cd bin/
-bash: cd: bin/: No such file or directory
[ec2-user@ip-172-31-23-184 webapps]$ cd ..
[ec2-user@ip-172-31-23-184 apache-tomcat-9.0.115]$ cd bin/
[ec2-user@ip-172-31-23-184 bin]$ ./startup.sh
Using CATALINA_BASE:   /home/ec2-user/apache-tomcat-9.0.115
Using CATALINA_HOME:   /home/ec2-user/apache-tomcat-9.0.115
Using CATALINA_TMPDIR: /home/ec2-user/apache-tomcat-9.0.115/temp
Using JRE_HOME:        /usr
Using CLASSPATH:        /home/ec2-user/apache-tomcat-9.0.115/bin/bootstrap.jar:/home/ec2-user/apache-tomcat-9.0.115/bin/tomcat-juli.jar
Using CATALINA_OPTS:
Tomcat started.
[ec2-user@ip-172-31-23-184 bin]$
```

- Access Application
- Open browser:

http://<Tomcat_Public_IP>:8080/addressbook-2.0

- The Address Book Application UI should appear.



First Name	Last Name	Email
George	White	george@white.com
Daniel	Thompson	daniel@thompson.com
Timothy	Jones	timothy@jones.com
Peter	Wilson	peter@wilson.com
Dan	Robinson	dan@robinson.com
Dan	Davis	dan@davis.com
Olivia	Davis	olivia@davis.com
Dan	Smith	dan@smith.com
Daniel	Anderson	daniel@anderson.com
Alice	Thomas	alice@thomas.com
Linda	Harris	linda@harris.com
Daniel	Robinson	daniel@robinson.com
Mike	Young	mike@young.com
Umberto	Anderson	umberto@anderson.com
Scott	Thompson	scott@thompson.com
Rene	Martin	rene@martin.com
Lisa	Martin	lisa@martin.com

Final Workflow

